MILITARY SPECIFICATION

PLUG AND DISC ASSEMBLY, 8597489

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

This specification provides requirements and quality assurance provisions for the fabrication and packaging of a plug and disc assembly, 8597489 for use in 76MM, 90MM, 105MM and 120MM Tank Ammunition.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification For
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-60539 - Coatings, Anodic, Conventional, for Aluminum and Aluminum Alloys

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

8597489 - Plug and Disc Assembly

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. Materials, parts, surface finish, assemblies, and salt spray testing shall comply with the requirements specified in applicable drawings and specifications.

3.2 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.3 Workmanship

3.3.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity and any other metal defects.

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3.3.2 Threads. Threads shall be full and undamaged for the entire minimum length and shall be in accordance with the provisions of MIL-A-2550.

3.3.3 Surface finish. All surfaces shall be free of burrs.

3.3.4 Protective coating. Protective coating shall be undamaged and uniformly applied to all designated surfaces regardless of contour configuration.

3.3.5 Foreign matter. All parts and assemblies shall be free of chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspections shall be conducted on this item:

a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a First Article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The First Article sample shall consist of the items in sample quantities as listed below:

Part Description	Drawing	Quantity
Plug	8597488	20
Disc	8595465	20
Plug and Disc Assembly	8597489	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70465 (AR)

PARAGRAPH	TITLE: Plug and Disc Assembly		SHEET 1 OF 1		DRAWING NUMBER: See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Plug - 8597488				
	a. Examination for defects	20		3.1	4.4.2.1
	b. Salt spray test	5		3.1	See Note
	Disc - 8595465				
	a. Examination for defects	20		3.1	4.4.2.2
	Plug and Disc Assembly 8597489				
	a. Examination for defects	10		3.1	4.4.2.3

NOTES: The samples shall be subjected to the salt spray test in accordance with MIL-C-60539 for 240 hours. If any sample fails this test, the first article shall be rejected.

DPSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plan. Unless otherwise specified in the Classification of Defects and Tests Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70465 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Plug	1 OF 2	8597488
			NEXT HIGHER ASSEMBLY 8597489

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Protective coating inadequate			3.3.4	4.5.1
102	Salt spray test			3.1	4.5.1.1
103	True position of inside diameter with major diameter of thread		0.40%	3.1	Gage
104	Pitch diameter of thread, min.		0.40%	3.1	Gage
105	Major diameter of thread, min.		0.40%	3.1	Gage
106	Diameter of hole		0.40%	3.1	Gage
107	Thickness of body		0.40%	3.1	Gage
108	Diameter of disc seat		0.40%	3.1	Gage
109	Metal defective		0.40%	3.3.1	Visual
110	Threads damaged or not full		0.40%	3.3.2	Visual
MINOR					
201	Pitch diameter of thread, max.		0.65%	3.1	Gage
202	Diameter of wrench holes		0.65%	3.1	Gage
203	Location of wrench holes		0.65%	3.1	Gage
204	Depth of wrench holes		0.65%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70465 (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.1	Plug		8597488
			NEXT HIGHER ASSEMBLY 8597489

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
205	Diameter of boss		0.65%	3.1	Gage
206	Depth of boss		0.65%	3.1	Gage
207	Depth of disc seat		0.65%	3.1	Gage
208	Radii or chamfers missing or incorrect		0.65%	3.1	Gage/Visual
209	Surface finish		0.65%	3.3.3	Gage/Visual
210	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70465 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Disc			8595465
				NEXT HIGHER ASSEMBLY 8597489

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Diameter		0.40%	3.1	Gage
102	Thickness		0.40%	3.1	Gage
103	Metal defective		0.40%	3.3.1	Visual
MINOR					
201	Burr		0.65%	3.3.3	Visual
202	Foreign matter		0.65%	3.3.5	Visual

NOTES:

DPSMC-OA ( ) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 w[illegible] may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8597489
4.4.2.3	Plug and Disc Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Improper assembly		0.40%	3.1	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

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4.4.3 Inspection equipment. Inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Protective coating. Procedures for testing, the cleaning of components, and the application of protective coatings shall be in accordance with he criteria for testing as specified on the applicable component drawings and in the specification referenced thereon. If a salt spray is required on the applicable specification, the salt spray test shall be accomplished on components or assemblies which have been manufactured in accordance with drawing requirements.

4.5.1.1 Salt spray test - plug. The salt spray test shall be accomplished in accordance with MIL-C-60539. Five samples shall be selected during first article and submitted to the salt spray test for 240 hours. Two samples for every 5000 plugs shall be selected during regular production (each lot) for the salt spray test. These samples shall be exposed for 120 hours, beginning with the first lot produced and continuing until three consecutive lots have complied with requirements. After three consecutive lots have been accepted, the samples of all subsequent lots shall be exposed for 72 hours. Should any lot fail the applicable requirements, the salt spray samples of the following lots shall be exposed for 120 hours until three consecutive lots have been accepted.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in MIL-STD-1169, Level B.

5.2 Packing and marking. Packing and marking shall be as specified in MIL-STD-1169.

5.3 Shipping. When assemblies from more than one lot are shipped as a car load, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

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6. NOTES

6.1 Intended use. Plug and disc assemblies are intended to be used to retain and protect the M13 Tracer on 76MM, 90MM, 105MM and 120MM Tank Ammunition.

6.2 Ordering data. See MIL-A-48078.

a. Data cards. Data cards shall be prepared for each lot in accordance with MIL-STD-1167.

b. Delivery of data cards to the technical agency. The following address shall be added to those listed in the contract (form DD 1423) for the delivery of one copy of the Ammunition Data Cards to the technical agency:

Commander
U.S. Army Armament Research and Development Center
ATTN: DRSMC-QAT-A(D)
Dover, NJ 07801

6.3 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, characteristics may be measured or gaged to determine acceptability.

6.4 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equiment designs, as required, to:

Commander
U.S. Army Armament Research and Development Center
ATTN: DRSMC-QAT-I(D)
Dover, NJ 07801

Custodian:
Army-AR

Preparing Actvity:
Army-AR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70465	PLUG AND DISC ASSEMBLY 8597489

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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